MEMORY DESIGN LAB

Memory Under Pressure: How Fixed-State Associative Memory Trades Memory Growth for Interference

TOPIC: Associative Memory & Fast Weights · TRACK: Pathway Track

SUBMISSION BRIEFING · DETERMINISTIC RESEARCH ARTIFACT

CENTRAL FALSIFIABLE SCIENTIFIC CLAIM

"A fixed-size linear associative memory can process a stream of arbitrary length without allocating a new memory slot for every association, but retrieval error can increase when stored key vectors overlap because other associations contribute cross-talk to the queried memory."

1. THE REAL ENGINEERING PROBLEM 01

Modern streaming agents process unbounded information (preferences, sensor feeds, customer states), yet cannot retain unbounded history in explicit representations. Standard Transformers maintain explicit key-value caches that grow linearly ($O(N)$) with sequence length, creating prohibitive latency and footprint. Conversely, bounded associative memories compress arbitrary streams into a fixed-capacity matrix state ($M \in \mathbb{R}^{d \times d}$), ($O(1)$) inference cost). This compression changes the engineering challenge: **a bounded state does not eliminate memory degradation; it trades memory state growth for retrieval interference**.

2. THE MATHEMATICAL MECHANISM 02

Consider unit-normalized key cues ($k_t \in \mathbb{R}^d$) ($\|k_t\|=1$), values ($v_t \in \mathbb{R}^d$), and associative state ($M_t \in \mathbb{R}^{d \times d}$). The baseline additive Hebbian write updates memory via rank-1 outer products: ($M_t = M_{t-1} + v_t k_t^\top$). Querying memory with ($q = k_j$) yields linear retrieval ($\hat{v}_j = M_t k_j$). Expanding across (N) stored associations produces the exact additive decomposition:

$$\hat{v}_j = Mk_j = v_j + \sum_{i \neq j} v_i (k_i^\top k_j)$$

Decomposition Insight: The first term is the isolated target memory. The second term is cross-talk: energy contributed by every other stored memory whose key has non-zero projection ($k_i^\top k_j \neq 0$) onto the query. Cross-talk is driven by cue geometry, not simply stream duration.

3. INTERACTIVE SUBSTRATE: MEMORY DESIGN LAB 03

To make this failure mode tangible, we engineered *Memory Design Lab*, an interactive client-side web application operating without cloud or backend dependencies. It provides two operational modes:

Mode 1 (Explore the Science): Learners adjust state dimension ($d \in \{4,8,16,32\}$), association count ($N \in [1,64]$), correlation parameter ($\rho \in [0, 0.9]$) via $k_i = \mathrm{normalize}(\sqrt{\rho}\, s + \sqrt{1-\rho}\, r_i)$, and rule (Hebbian vs. Delta). Computes live SVG heatmaps, error sweeps, and exact KV baselines in browser.

Mode 2 (Scenario Stress-Test Lab): Engineers test bounded memory on realistic synthetic tasks (AI Coding Agent Context, Customer Support Tickets, IoT Sensor Stream) to observe which specific facts survive cross-talk before shipping.

4. MEASURED EXPERIMENTAL EVIDENCE 04

All metrics are evaluated live via a seeded Mulberry32 PRNG (tested in 95 automated unit tests):

Condition	Config (ρ)	Mean Overlap	Mean Cosine	Retrieval MSE
Low Overlap	0.10	0.2690	0.6783	0.1563
High Overlap	0.80	0.8031	0.3607	**1.0890**
Intervention	+0.70	+0.5341	-0.3176	**+0.9327 (6.97×)**

Direct Empirical Validation: Holding seed (42), state dimension ($d=8$), association count ($N=12$), values, and update rule identical, increasing key overlap produced a **6.97× increase in measured retrieval MSE**. *(Observed in this seeded experiment; not a universal law.)* In a fixed (8×8) capacity sweep, scaling ($N=2 \to 64$) drove MSE from 0.0075 to 1.5896 (>210×), demonstrating capacity pressure in a fixed state.

5. MODERN RESEARCH CONTEXT 05

This toy system isolates the core design pressure of modern sequence models: **RetNet** [arXiv:2307.08621] derives recurrence-attention duals with ($O(1)$) inference memory; **GLA** [arXiv:2312.06635] utilizes data-dependent decay gates over matrix-valued states; and **Gated DeltaNet** [arXiv:2406.06484] combines gating with delta error-correction updates ($\Delta M = \beta(v_t - M_{t-1}k_t)k_t^\top$) to selectively erase stale memory.

6. THE DRAGON HATCHLING (BDH) ANCHOR 06

BDH (arXiv:2509.26507, *The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain*, Kosowski et al., Sept 30, 2025) provides the architectural anchor for inference-time working memory:

1. **Synaptic Plasticity:** BDH implements working memory during inference via synaptic plasticity and local Hebbian learning ($\Delta \text{synapse}_{ij} \propto \text{pre}_j \times \text{post}_i$), dynamically strengthening synapses as concepts are processed.
2. **Sparse Positive State:** Activations are strictly non-negative and highly sparse, unlike dense unconstrained outer-product memories.
3. **Educational Boundary:** Our matrix memory is an *educational toy abstraction*, not an implementation of BDH. While our simulator uses dense matrix multiplication, BDH operates over a scale-free particle graph. The mechanisms are conceptually related but architecturally distinct.

7. THE BDH-CQ REASONING BRIDGE 07

BDH-CQ [arXiv:2608.09888] extends the synaptic family by maintaining continuous recurrent working memory during inference, followed by iterative multi-step reasoning in a latent state space without verbalizing chain-of-thought tokens.

8. ARCHITECTURAL TRADE-OFFS 08

Approach	Memory State	Primary Advantage	Core Trade-off
Exact KV Store	List ($\{(k_i, v_i)\}$)	Zero retrieval error	State scales ($O(N)$)
Additive State	Matrix ($M \in \mathbb{R}^{d \times d}$)	Bounded ($O(1)$) state	Cross-talk interference
Delta Update	Corrected matrix	Corrects recent cues	Complex dynamics
BDH Synaptic	Particle synapses	Native working memory	Specialized graph runtime

9. SCIENTIFIC BOUNDARIES & OPEN QUESTIONS 09

Honest Boundary: This experiment uses synthetic vectors and a minimal linear outer-product mechanism. It models the behavior of this specified toy system, not the full capability of scaled recurrent language models.

Open Research Question: How effectively can richer update rules, data-dependent decay gating, activation sparsity, and learned representation geometry preserve working memory in bounded state without introducing unacceptable cross-talk?

10. PRIMARY LITERATURE REFERENCES 10

[1] BDH: Kosowski et al., *Dragon Hatchling*, arXiv:2509.26507 (2025).
[2] BDH-CQ: Engdahl et al., *Recurrent Latent Reasoning*, arXiv:2608.09888 (2026).
[3] RetNet: Sun et al., *Retentive Networks*, arXiv:2307.08621 (2023).
[4] GLA: Yang et al., *Gated Linear Attention*, arXiv:2312.06635 (2023).
[5] Gated DeltaNet: Yang et al., *Delta Rule Transformers*, arXiv:2406.06484 (2024).